

How the detector works, and why

A slower walkthrough for someone who does not write code. Every technical name is defined. Every design choice has a “why.”

Val acierto 97.2% (69 / 71) · set completo 96.6% · 22 ms en vivo · no es el set oculto de jueces

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1

The question

A bank’s AI voice agent is already on the line. Someone calls in. That incoming voice is either a real person or a machine stack — speech recognition + language model + synthetic voice — improvising through the same script. We answer one bit: **is the incoming caller synthetic?**

SYNTHETIC CALLER

Not a recording played down the line. A live loop: hear the agent (ASR), invent a reply (LLM), speak it (TTS). That is why we measure *conversation timing* as well as the sound of the voice. A replay cannot wait, recover, or barge in the same way. A stack can — but it waits more uniformly.

Why we never try to identify the person

Human callers volunteered, were told the call was recorded, and used invented personal data. The job is “human vs machine,” not “who is this.” The model never sees a name, an account number, or a speaker ID.

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What arrives at the endpoint

Judges hit one URL for the whole visit: `POST /detect`. The body is JSON. The important field is `audio_base64` — the entire stereo WAV encoded as text so it can travel in that JSON.

BASE64

A way to write binary (the WAV file) using only letters, numbers, and a few symbols. Email and HTTP do the same thing. We decode it back into bytes, then into samples. If that decode fails, we return HTTP 400, not a guess.

Optional fields `sample_rate` and `channels` are metadata. We do not trust them for the split. We read the WAV header ourselves. We accept 8 kHz (the challenge) and 16 kHz (local experiments). Anything else is rejected.

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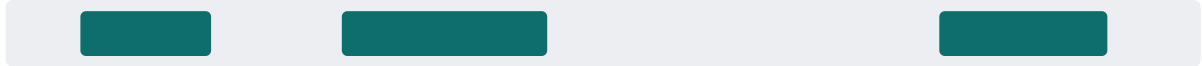
How we split channel 0 and channel 1 — and how we know which is which

This is the most common confusion, so it is worth saying twice: **we do not infer who is the caller**. We do not listen and decide “this side sounds like a customer.” We do not train a speaker-identification model. We do not pick the louder track. The challenge wrote the rule down. We follow it.

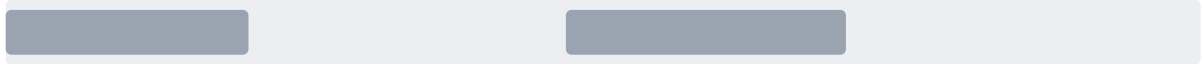
What “stereo” means here

A mono file is one stream of numbers: air pressure, many times per second. A stereo file is **two** streams recorded in lockstep, stored as two columns in the same WAV. In music, those columns are “left speaker” and “right speaker.” In this challenge they are two *roles* on a phone call.

WAV column 0 · caller · the only voice we classify



WAV column 1 · bank agent · context only



Same clock. Overlap is allowed. We never mix the two columns back together.

DEMUX

Short for demultiplex: pull the two columns apart. In code that is `data[:, 0] → caller` and `data[:, 1] → agent`. If the file has fewer than two channels, we refuse it (`NotStereoError`). We do not invent a second channel.

Why we “know” channel 0 is the caller

Because the organizers said so, in the brief and in the dataset: **channel 0 = incoming caller (classification target)**, **channel 1 = agent**. Every train and val WAV was built that way. The live judge audio is built the same way. That is a *contract*, like “JSON field `is_synthetic` is a boolean.” It is not a fact we discovered from the waveform.

What we are not doing

We are not running diarization (“who spoke when, as people”). We already have two wires. We are not comparing voices to decide roles. If a file arrived with the wires swapped, we would classify the agent and treat the caller as context — and we would be wrong. The system trusts the layout the same way a bank trusts “this RJ45 is the WAN port.”

Why we still keep the agent channel

The organizers’ hint: you get both channels for a reason. The caller track is the voice we judge. The agent track is the *stimulus* — when the bank bot stops, interrupts, or talks over the caller. Humans and machines recover from that turbulence differently. Without channel 1 we could

still measure how messy the caller sounds. We could not measure how long they wait after the agent yields the floor.

That wait is the strongest behavioural clue we ship (Cohen's $d \approx 1.90$). It only exists because we know *when the agent stopped*, which we only know because we segmented channel 1.

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Why the audio is so thin

Challenge audio is telephony-grade: 8 kHz, 16-bit PCM. A studio podcast is often 44.1 or 48 kHz. Eight thousand samples per second can represent frequencies only up to 4 kHz (the Nyquist limit). Human speech still fits. Breath, fine pitch jitter, and high formants mostly do not.

8 KHZ / PCM / NYQUIST

PCM means the file stores raw samples, not a compressed codec like MP3. **Nyquist** says you need two samples per cycle of the highest frequency you want to keep. At 8 kHz you keep up to 4 kHz — enough for vowels and most consonants on a phone line, not enough for “I heard them inhale.”

Why this killed several clever ideas

We looked for audible breaths and room echo of the agent leaking into the caller mic. On this band they scored $d = 0.00$. The ideas are not stupid. The medium hid them. That is why we kept loudness *variation* and timing, not “biological” micro-cues.

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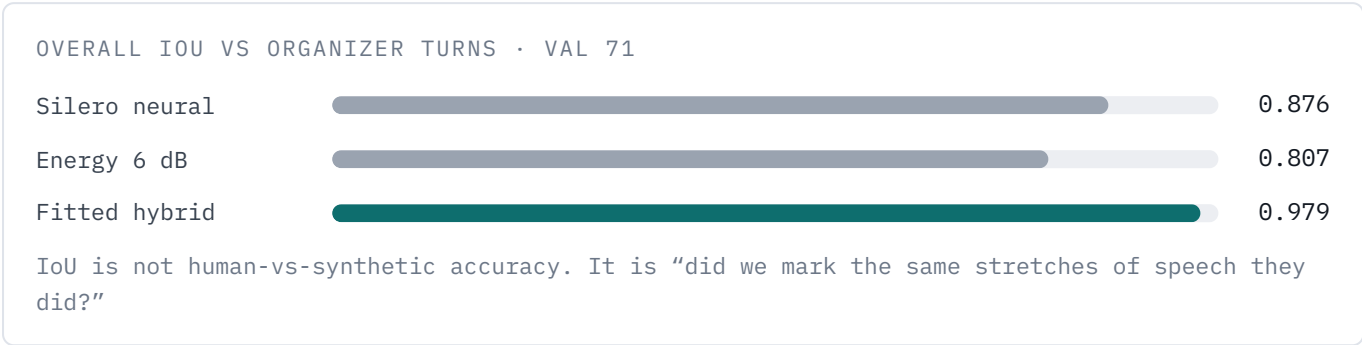
Finding speech (VAD)

At judging we receive audio only. The organizer files `turns/*.json` exist for train and val so we can score ourselves. They will not be in the request. We must rebuild “who spoke when” from the waveform.

VAD – VOICE ACTIVITY DETECTION

A detector that answers, for each tiny slice of time: is this channel speaking, or not? We run it *independently* on channel 0 and channel 1. Then we can see overlap (both true), silence (both false), and floor switches (one stops, the other starts).

We tried a neural VAD (Silero) and a simple energy VAD (is this frame louder than the quiet floor + 14 dB?). The organizer labels behave like energy, not like Silero. Overlaying Silero on fitted energy *hurt* caller overlap. So the shipped backend is **hybrid**: use fitted energy; call Silero only if energy hears nothing at all.



IOU – INTERSECTION OVER UNION

Imagine two highlighters on a timeline: ours and the organizer’s. Intersection is the time both marked as speech. Union is the time either marked. IoU = intersection ÷ union. 1.0 is a perfect match. We need this high so the features we compute at serve time look like the features we trained on. If VAD drifts, wait times and “messiness” drift with it.

Why 14 dB, not 6 dB

Energy VAD says “speech” when a 25 ms frame is more than N decibels above a quiet-floor estimate (the 12th percentile of frame energy). 6 dB almost never missed speech but painted a lot of noise as speech (IoU ~0.80). 14 dB sits on a plateau with 13–15 dB (IoU ~0.95). 16 dB starts to miss. We picked the middle of the plateau, confirmed on val, not a one-call spike.

After the mask we drop blobs shorter than 120 ms and glue gaps shorter than 280 ms. That matches how the organizer JSON merges pauses.

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The turn ledger

Once both channels have speech segments, we lay them on one timeline. That book of events is the **turn ledger**.

EVENT	WHAT IT MEANS	WHY WE CARE
Speech	Only one side is talking	Slice the caller’s audio for acoustic features

EVENT	WHAT IT MEANS	WHY WE CARE
Overlap	Both sides talking at once	Turbulence; humans barge in more
Silence	Neither side	Dead air; also the gap before a reply
Floor switch	Agent stops, then caller starts (or the reverse)	The wait we measure

FLOOR-SWITCH LATENCY

How many seconds pass after the agent's speech ends before the caller's speech begins, counting only positive waits (the caller did not already talk over the agent). Val humans sit near 1.22 s median. Val synthetics sit near 2.56 s. A stack is hear → think → speak. It cannot start until it detects the agent finished. A person predicts the end of the sentence.

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Two kinds of evidence

The organizers named three attack surfaces. We ship two. Semantic (what was said) is coded but not fused: no transcripts on disk, and seconds of extra latency.

Acoustic — the voice itself

During caller speech segments we measure how much the sound *wanders*: loudness (RMS), buzziness (zero-crossing rate), brightness (spectral centroid), tone-vs-noise (spectral flatness), peakiness (crest factor). Humans wander. Vcoders hold a steadier operating point.

Why we dropped average loudness

caller_rms_mean is the strongest separator in this zip: synthetics are louder (train means 0.15 vs 0.07). That is probably how they were injected — a fixed digital level — not a law of fake voices. The hidden set may not repeat it. We kept caller_rms_cv (how much loudness *varies*) instead. Hybrid stacked accuracy did not fall when we dropped the mean.

CV, STD, RMS, ZCR – THE ACOUSTIC WORDS

RMS is a standard way to say “how loud” (root-mean-square of the samples). **Std** is how spread out a list of those loudnesses is. **CV** (coefficient of variation) is $\text{std} \div \text{mean}$: spread relative to typical size, so a quiet messy talker still counts as messy. **ZCR** counts how often the waveform crosses zero — a cheap “buzziness / noisiness” number.

Behavioural — the conversation

Three numbers: typical wait after the agent stops; how much of the call the *agent* is talking (humans leave more room); how inconsistently the caller restarts after the agent (recovery CV). Timing wins ties: the mixer weight is 2.41 for behavioural vs 1.65 for acoustic.

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How a “head” turns numbers into a probability

Each head is a **logistic regression**, not a deep net. That is a deliberate feasibility choice: a few weights in a JSON file, no GPU, readable.

1. Take the feature values (for example five acoustic numbers).
2. Standardize them: subtract the train mean, divide by the train std, so “loudness CV” and “brightness std” sit on the same scale.
3. Dot product with learned weights, plus a bias. That sum is a **logit**.
4. Pass the logit through a sigmoid: $1 / (1 + e^{-x})$. The result is a number between 0 and 1: **P(synthetic)** according to that head.

HEAD

One specialist model. Acoustic head sees only voice-texture features. Behavioural head sees only timing features. They can disagree. That disagreement is useful: it is how we know when the mixer is guessing.

On val, acoustic alone is 87.3% accurate (AUC 0.963). Behavioural alone is 93.0% (AUC 0.979). Neither is the system we ship. Together they catch different mistakes.

How the mixer combines the two opinions

Stacked fusion means a third logistic model whose only inputs are the two head probabilities. It learned, on out-of-fold train scores, to trust timing more than texture. We do not average 0.5/0.5. We do not pick the max. We let the data pick the weights.

Why “out of fold”

If you train the heads on the same rows you then feed to the mixer, the mixer memorizes “this train call’s acoustic score” instead of learning the relationship between the two scores. We train heads on 4/5 of train, score the held-out 1/5, rotate, then fit the mixer on those honest scores. That is stacked fusion with 5-fold OOF logits.

We also fitted a concatenated model (all eight features in one soup). It can look slightly better on some previews. We ship stacked because it is the honest live path, stays within 0.01 AUC, and does not chew collinear wait statistics plus the loudness trap.

Two safety rules after the mixer

These are not new classifiers. They are tiny, named exceptions we measured on train and confirmed on val.

UNSURE DISAGREEMENT

If the heads vote opposite sides of 0.5, *and* the mixer’s fused probability is within 0.10 of a coin flip, we take the sharper head (the one farther from 0.5). On val that saved one human and broke none. Applying “always trust the sharper head” on all 10 disagreements was a wash: save 2, break 2. So the band stays narrow.

TALKATIVE-QUIET GATE

A polite person can wait 1.4–2.0 s (bot-like) and keep tidy volume, yet still take many turns. If wait is in that band, `rms_cv` is below 0.45, and there are 24 or more caller turns, we flip the verdict toward human. That saved e2d538 and is why val moved from 95.8% to 97.2%. The gate is class-agnostic (wait + tidy energy). The expert only fires for the talkative subtype, so sparse quiet bots are not mass-pardoned.

What we send back

FIELD	WHAT IT IS	HOW IT IS MADE
is_synthetic	Required yes/no	True if $P(\text{synthetic}) \geq 0.5$
confidence	How sure we are of that yes/no	P if we said synthetic, else $1 - P$
views	Debug only	Per-head $P(\text{synthetic})$; judges may ignore

Why confidence is not $P(\text{synthetic})$

If $P = 0.12$ we say `is_synthetic: false` with confidence 0.88: “we are 88% sure this is a person.” The judge reconstructs P as confidence when the flag is true, and $1 - \text{confidence}$ when false. That lets them compute AUC and Brier without a third field.

Example: sure machine $P=0.87 \rightarrow \text{true} / 0.87$. Sure person $P=0.12 \rightarrow \text{false} / 0.88$. Coin flip $P=0.51 \rightarrow \text{true} / 0.51$.

How to read the scores

ACCURACY

Share of calls we get right at the 0.5 line. Val: $69/71 = 97.2\%$. Easy to say. Sensitive to class balance and to that one threshold.

AUC

Area under the ROC curve. Pick one human and one synthetic at random. How often does the synthetic get the higher P ? 0.5 is a coin. 1.0 is perfect ranking. Ours on val stacked is 0.997. This does not care about the 0.5 cutoff — only order.

TPR / TNR / BALANCED ACCURACY

TPR (true positive rate): of the synthetics, how many did we catch. Full set 98.0% (199/203). **TNR** (true negative rate): of the humans, how many did we clear. Full set 94.7% (142/150). **Balanced accuracy** is the average of those two, so a lopsided dataset cannot hide “we just always say synthetic.”

BRIER SCORE

How honest the probabilities are, not just the yes/no. Saying 0.99 on a coin-flip call is punished. The judge can compute this from the recovered $P(\text{synthetic})$.

COHEN'S D

How far apart two group means sit, in standard deviations. $|d| = 0.2$ small, 0.5 medium, 0.8 large. Our shipped clues sit between 1.2 and 1.9. Sign tells direction: wait is higher for synthetics; messiness is higher for humans.

SPEAKER-DISJOINT

No caller in train appears in val. Judging is a third group of people and voices. Memorizing “this polite human waits 1.75 s” will not transfer. That is why we froze on the last two val misses instead of adding a feature for them.

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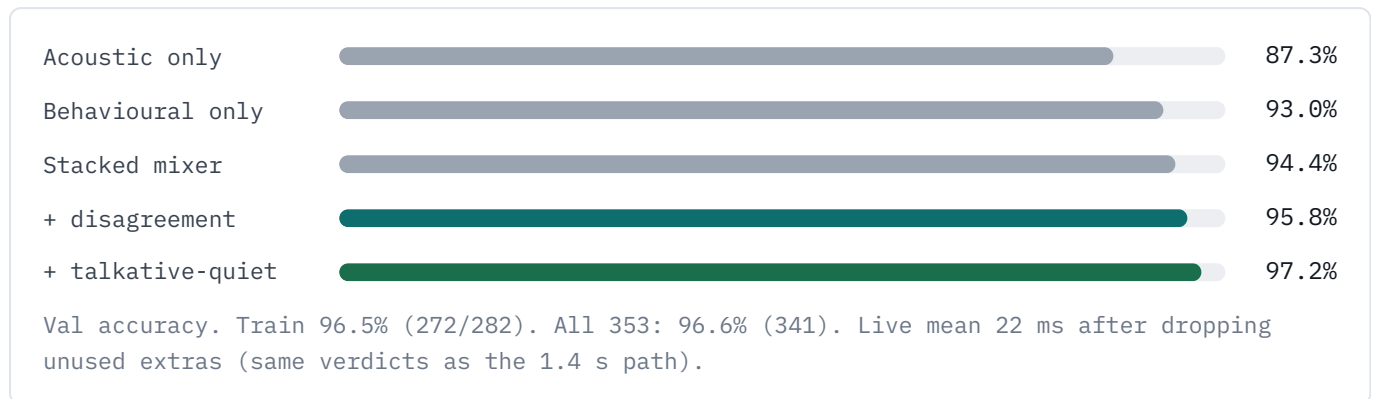
What we measured and threw away

We extracted 68 numeric features. Eight shipped. The unused ideas are part of the result: we know they failed on this medium.

IDEA	WHY IT SOUNDED GOOD	WHY IT DIED
Breath / inhalations	People inhale; bots may not	$d = 0.00$ at 8 kHz, short banking utterances
Room bleed / echo	A real phone should pick up the agent in the room	Call conditions vary; no stable split
Average loudness	Synths in this zip are louder	Injection trap; hidden set may not match
Formant volatility	Mouth shape should wander	Ranks high on val, points the wrong way; LPC at 8 kHz is unreliable
Pitch jitter	Vocal-fold micro-wobble	val $d \approx 0$
Live Whisper / semantics	Bots invent fake account facts	No transcripts; trap lines are often the agent's script; seconds of latency

The organizer line we followed: **depth beats breadth**. One signal done well outscores three that half-work.

Current numbers (labeled set)



Two val misses remain. 6971 is a careful quiet human both heads call synthetic. d4518 is a synthetic the acoustic head already has (0.87); the mixer trusted a human-like 1.74 s wait. We froze. Chasing both memorizes types that will reappear differently on the hidden set.

Glossary A–Z

8 kHz

Phone-line sample rate. Keeps frequencies up to 4 kHz. Hides breath and fine pitch.

Agent (channel 1)

The bank's AI voice. Never classified. Used as the stimulus the caller reacts to.

ASR + LLM + TTS

Hear, think, speak. The synthetic stack we are hunting. Not a replayed clip.

AUC

Ranking quality. Probability that a random synthetic scores higher than a random human.

Base64

Text encoding of the WAV so it can sit inside JSON.

Caller (channel 0)

The incoming voice. The only target. Assigned by the challenge layout, not inferred.

Cohen's d

Standardized gap between group means. $|d| > 0.8$ is large.

Confidence

Certainty in the yes/no label, not raw $P(\text{synthetic})$. Judge inverts it when the label is false.

CV (coefficient of variation)

Spread $\div$ typical size. Lets a quiet messy talker still look messy.

Demux

Split stereo columns: $[:,0]$ caller, $[:,1]$ agent.

Diarization

Inferring who spoke from a mixed recording. We do not do this; we already have two wires.

Feature

One number computed from a call (wait time, loudness CV, ...).

Floor switch

One side stops; the other starts. The wait after the agent stops is our strongest timing clue.

Head

One logistic specialist (acoustic or behavioural).

Hybrid VAD

Fitted energy detector; Silero only if energy returns no segments.

Injection artifact

A pattern true in this zip (synths louder) that may be a side effect of how files were made.

IoU

Overlap quality of our speech segments vs organizer segments. High IoU keeps train and serve aligned.

Logistic regression

Weighted sum of standardized features, then a sigmoid $\rightarrow$ probability. No deep net in production.

Logit

The number before the sigmoid. Positive leans synthetic.

Overfit

Memorizing this dataset's leftovers so you fail on new speakers. Why we froze on two misses.

$P(\text{synthetic})$

Our estimated probability the caller is a machine. Verdict is yes at 0.5.

PCM

Uncompressed samples in the WAV.

RMS

Root-mean-square energy; a loudness number.

Sigmoid

S-shaped curve that squashes any real number into (0, 1).

Speaker-disjoint

Train, val, and hidden set share no callers.

Stacked fusion

A third logistic model on the two head probabilities, trained on out-of-fold scores.

Stereo

Two synchronized tracks in one file. Here: roles, not left/right music.

Turn ledger

Timeline of speech, overlap, and silence on both channels.

VAD

Voice activity detection; speech vs silence per channel.

Vocoder / TTS

The machine that renders the synthetic voice. Tends to hold texture steadier than a throat.

ZCR

Zero-crossing rate; a cheap buzziness measure.

Labeled-set figures only. Hidden judging audio is unseen. Channel roles are a challenge contract: ch0 caller, ch1 agent. We do not infer them. Print this page to PDF (File → Print → Save as PDF) or use the generated PDF in docs/.